Sparse Autoencoders Learn Monosemantic Features in Vision-Language Models

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Citation 116

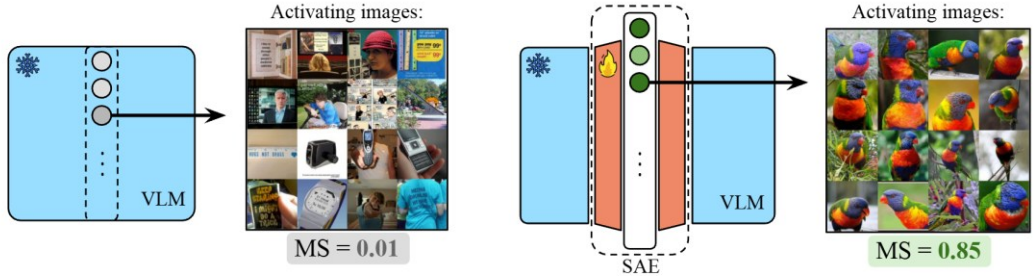


Figure 1: Sparse Autoencoder (SAE) in VLM (e.g. CLIP): Top activating images of a neuron in a pretrained VLM layer are polysemantic (left), and those of a neuron in a sparse latent of SAE trained to reconstruct the same layer are monosemantic (right), according to MonoSemanticity score (MS).

Department of Computer Engineering and Artificial Intelligence
MAI Lab.

YS Park

국립부경대학교 PUKYONG NATIONAL UNIVERSITY

Medical AI laboratory

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Introduction

- CLIP과 SigLIP 같은 VLM들이 널리 적용되고 있으나, 아직 내부 구조를 이해하는 것은 아직 제한적임.
- Sparse AutoEncoders (SAEs)는 특정 모델의 내부 표현을 탐색하기에 효과적인 접근 방법이며, LLMs 내부 접근에는 많이 사용되고 있음.
- VLM을 위해서는 interpretable classification이나 cross-model concept alignment같은 방법에 국한되고 있음.
- Contribution 1: VLM이 널리 사용되는 것에 비해 neuron-level에서 SAE를 검증하는 metric의 부재를 **MonoSemanticity score**로 해결함.
- Contribution 2: SAE 레이어에서 개념들이 잘 분리되는 특성을 활용하여 neuron activation에 개입하고 MLLM의 출력을 조절하여 발견된 개념을 삽입하거나 억제함을 보임.

Related Study

- **SAEs**
  - **SAEs**는 LLMs과 VLMs의 내부 표현을 이해하기 위해 사용됨.
  - Batch TopK, JumpReLU, Matryoshka representation learning 등이 연구되어 사용되고 있음.

- **VLMs**
  - **VLMs**은 CLIP의 등장과 함께 급 부상한 패러다임임.
  - 다양한 벤치마크에 있어서 강력한 성능을 보이고 있음.
  - 이와 동시에 VLMs의 신뢰가능성을 논하는 연구도 활발히 수행되고 있음.
  - 기존연구에서는 모델 내부의 표현을 텍스트 기반으로 분해하거나 개별 해석 가능한 뉴런을 찾으려는 기존 연구들이 있었음.

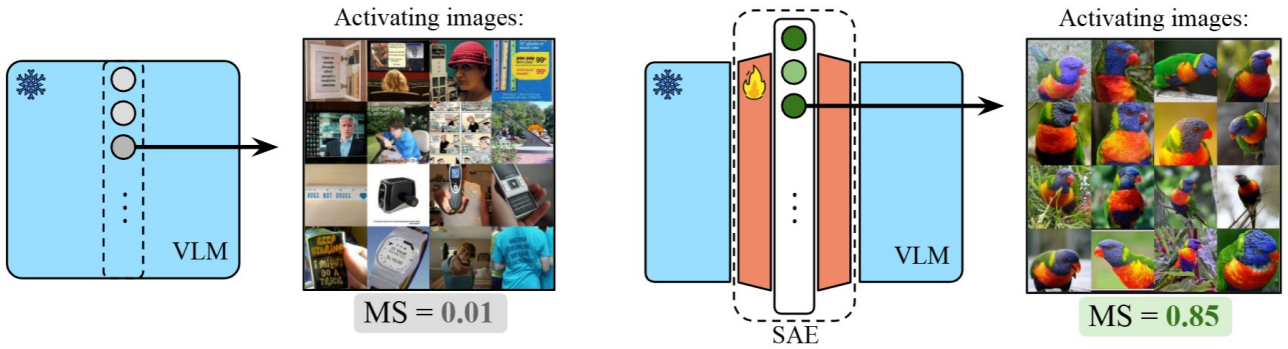


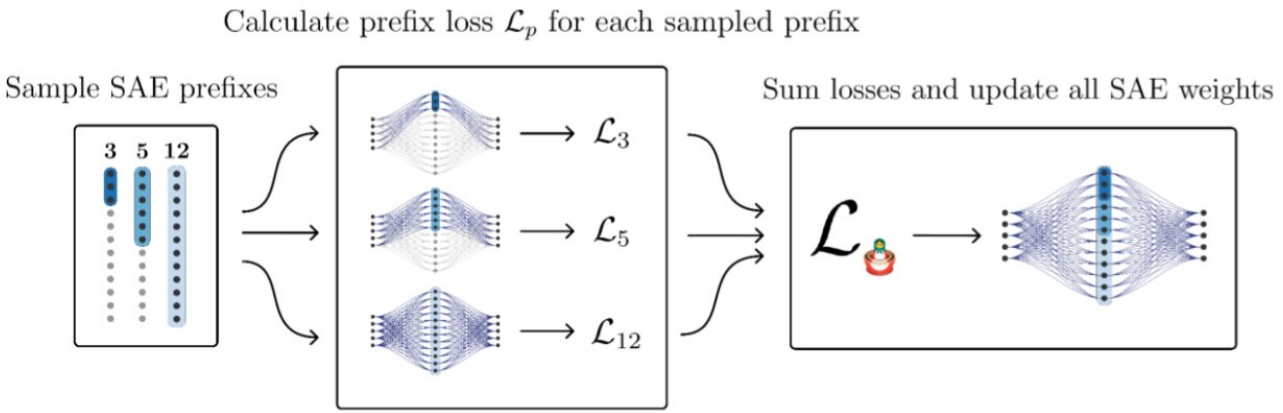
Figure 1: Sparse Autoencoder (SAE) in VLM (e.g. CLIP): Top activating images of a neuron in a pretrained VLM layer are polysemantic (left), and those of a neuron in a sparse latent of SAE trained to reconstruct the same layer are monosemantic (right), according to MonoSemanticity score (MS).

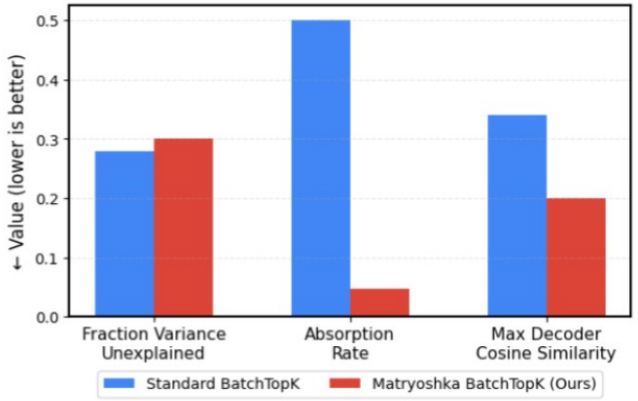
Background and Formulation of SAEs

- SAE 기본 구성:
  - 구성: $\mathrm{W_{enc}} \in \mathbb{R}^{d \times w}, \mathrm{W_{dec}} \in \mathbb{R}^{w \times d}, \mathrm{b} \in \mathbb{R}^{d}, \sigma = \mathbb{R}^{w} \rightarrow \mathbb{R}^{w}$
  - 입력: $\mathrm{v} \in \mathbb{R}^{d}, \mathrm{v} := f_l(x)$
  - 출력: $\phi(\mathrm{v}) := \sigma\left(\mathrm{W_{enc}^{\top}}(\mathrm{v} - \mathrm{b})\right), \psi(\mathrm{v}) := W_{dec}^{\top}\mathrm{v} + \mathrm{b}, \hat{\mathrm{v}} := \psi(\phi(\mathrm{v}))$
  - Loss: $L(\mathrm{v}) := R(\mathrm{v}) + \lambda S(\mathrm{v})$
    - $\mathrm{R}(\mathrm{v}) := \| \mathrm{v} - \hat{\mathrm{v}} \|_2^2$
    - $\mathrm{S}(\mathrm{v}) := \| \phi(\mathrm{v}) \|_1$

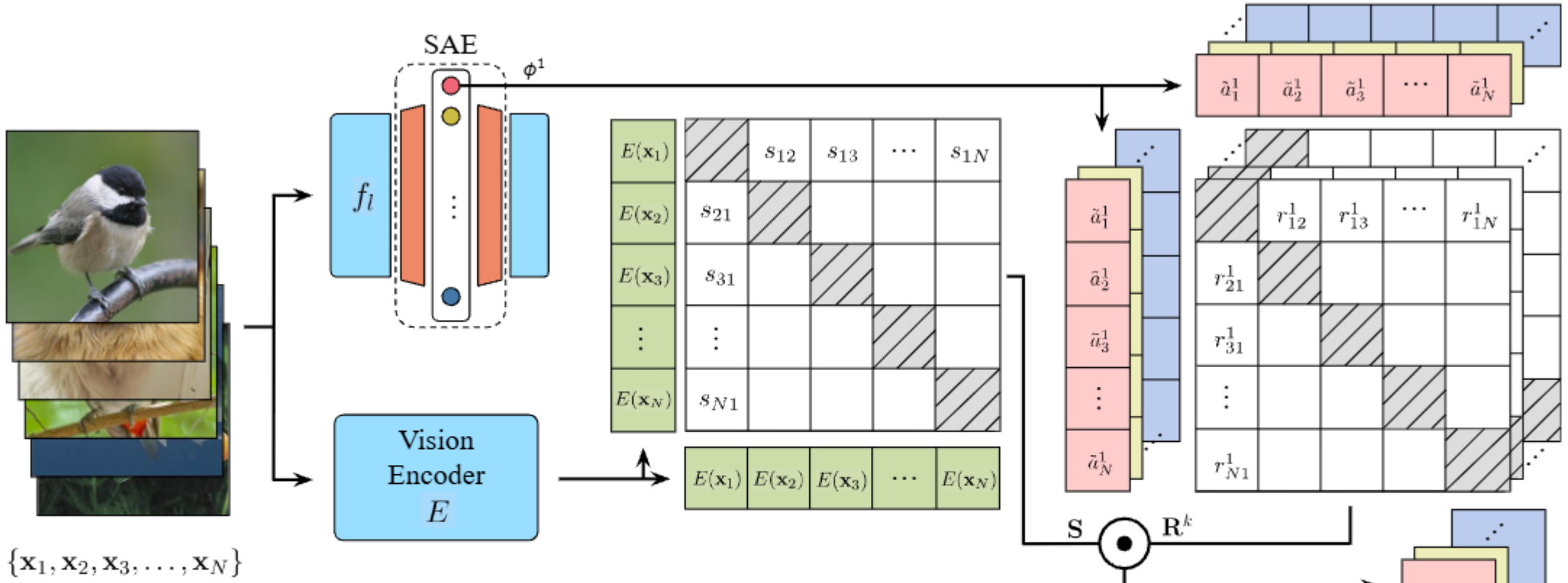
- Matryoshka SAE:

$$R(v) := \sum_{m \in \mathrm{M}} \| \mathrm{v} - W_{dec}^{\top}\phi^{1:m}(\mathrm{v}) \|_2^2$$





Monosemanticity Score



(a) Extracting embeddings and activations (b) Pairwise similarity matrix (c) Activation-weighted average

Figure 2: Computation of our MonoSemanticity score (MS). (a) Embeddings and activations are extracted for a set of images (b) to compute the pairwise embedding similarities and pairwise neuron activations. (c) MS is the average of embedding similarities weighted by the neuron activations.

Monosemanticity Score

$$L = \{x_n \in \mathbb{X}\}_{n=1}^{N}, E \coloneqq pretrained\ image\ encoder$$

$$s_{nm} = \frac{E(x_n) \cdot E(x_m)}{|E(x_n)||E(x_m)|} \in [-1, 1]$$

$$\mathrm{v}_n = f_l(x_n), a_n^k = \phi_{k(v_n)}, \{a^k = [a_n^k]_n \in \mathbb{R}^N\}_{k=1}^{\omega}$$

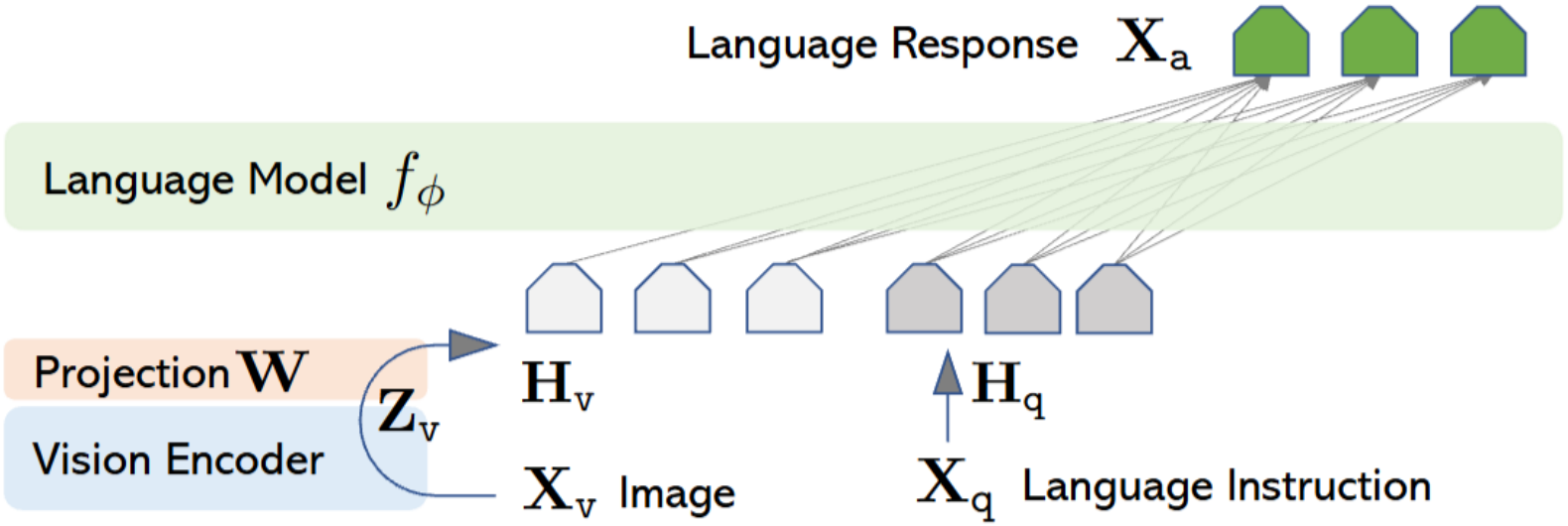
$$\tilde{\mathrm{a}}_n^k = \frac{a_n^k - \min_{n'} a_{n'}^k}{\max_{n'} a_{n'}^k - \min_{n'} a_{n'}^k}$$

$$r_{nm}^k = \tilde{\mathrm{a}}_n^k \tilde{\mathrm{a}}_m^k \in [0, 1]$$

$$MS^k = \frac{\Sigma_{1 \le n < m \le N}\, r_{nm}^k s_{nm}}{\Sigma_{1 \le n < m \le N}\, r_{nm}^k} \in [-1, 1]$$

Steering MLLMs with Vision SAEs

- Monosemantic한 neuro을 찾는 것은 interpretability에서만 중요한 것은 아님.
- 본 논문에서는 모델 본체를 파인튜닝할 필요 없이, 비전 인코더의 표현을 SAE의 '특정 단어/개념 담당 뉴런' 다이얼만 손으로 돌려(α값 조절) 모델의 시각적 사고방식을 마음대로 조종(steering)함.



Steering MLLMs with Vision SAEs

구분	세부 내용
대상 모델	• CLIP ViT-L/14-336px • SigLIP SoViT-400m/14-384px • AIMv2 L/14-224px • WebSSL MAE-300m/14-224px
학습 데이터셋	ImageNet 이미지에 대한 모델의 사전 추출 활성화 벡터 (Activation vectors)
추출 토큰 및 레이어	• CLIP: Residual stream의 CLS 토큰 (레이어 $l \in \{11, 17, 22, 23\}$) 또는 최종 프로젝션 레이어 출력 • Steering 실험: CLIP 레이어 $l = 22$의 이미지당 무작위 토큰 임베딩 2개 사용 • 기타 인코더: 최종 레이어 CLS 토큰 (CLS 토큰이 없는 경우 무작위 토큰 임베딩 2개)
SAE 아키텍처	• BatchTopK SAE • Matryoshka BatchTopK SAE
Matryoshka 그룹 (M)	$M = \{0.0625\omega, 0.1875\omega, 0.4375\omega, \omega\}$ (각 하위 레벨로 갈수록 추가되는 뉴런 수가 약 2배씩 증가)
희소도 제약 (K)	$K = 20$ (BatchTopK에서 non-zero 잠재 뉴런 최대 개수 고정)
확장 계수 (ϵ)	$\epsilon \in \{1, 2, 4, 8, 16, 64\}$ ($\omega := d \times \epsilon$)
최적화 기법 (Optimizer)	Adam Optimizer
학습 스텝 및 배치 크기	• 학습 스텝: 10^5 steps • 미니배치 크기: 4,096
초기 학습률 (Learning Rate)	$\frac{16}{125\sqrt{\omega}}$
평가 지표 (SAE Metrics)	• 복원 품질: R^2 • 활성화 희소도: L_0 norm
MS 측정용 이미지 인코더 (E)	DINOv2 ViT-B (기본 설정, 부록에서 CLIP ViT-B 등으로 추가 비교)
하드웨어 환경	단일 NVIDIA A100 GPU 1대

Evaluating Interpretability of VLM Neurons

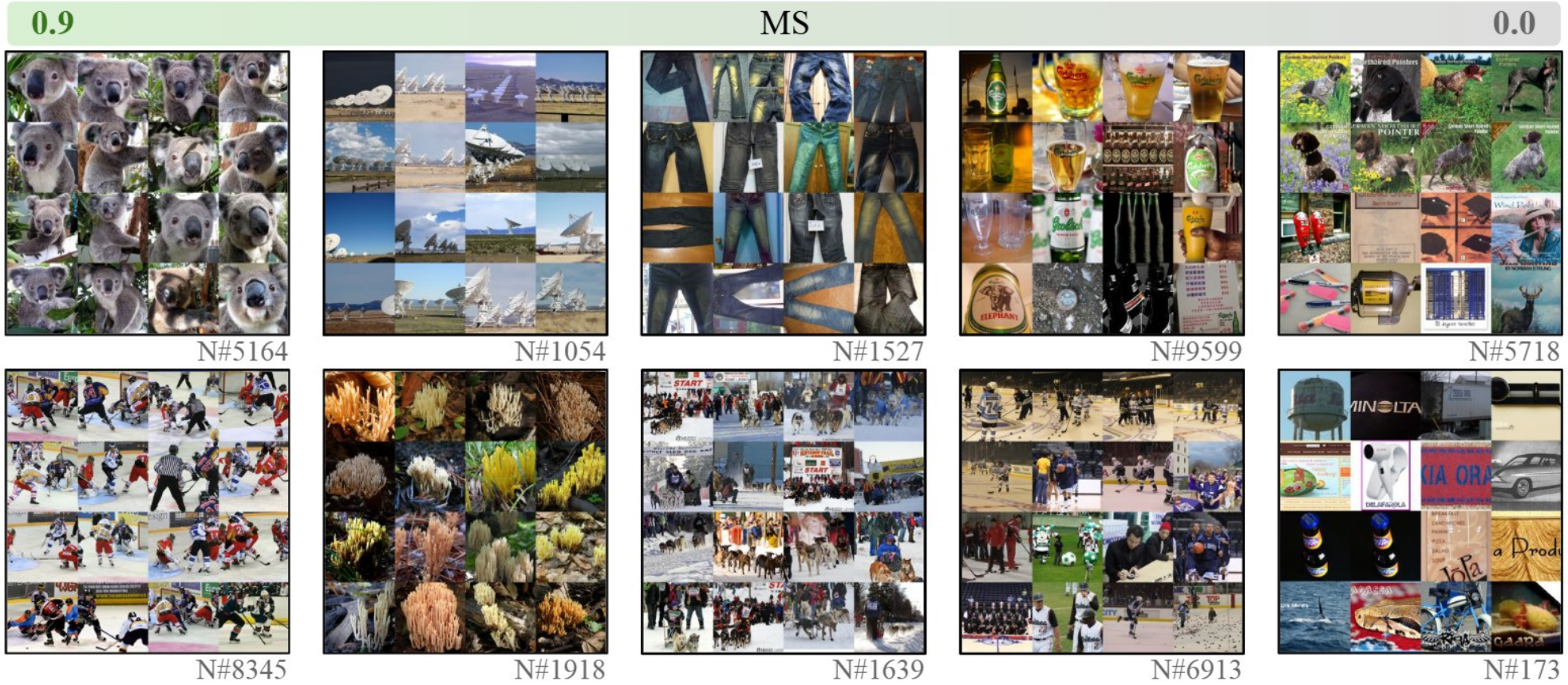


Figure 3: Top activating images of neurons with MonoSemanticity (MS) scores ranging from high (left) to low (right). Higher scores correlate with more similar images, reflecting monosemanticity.

Evaluating Interpretability of VLM Neurons

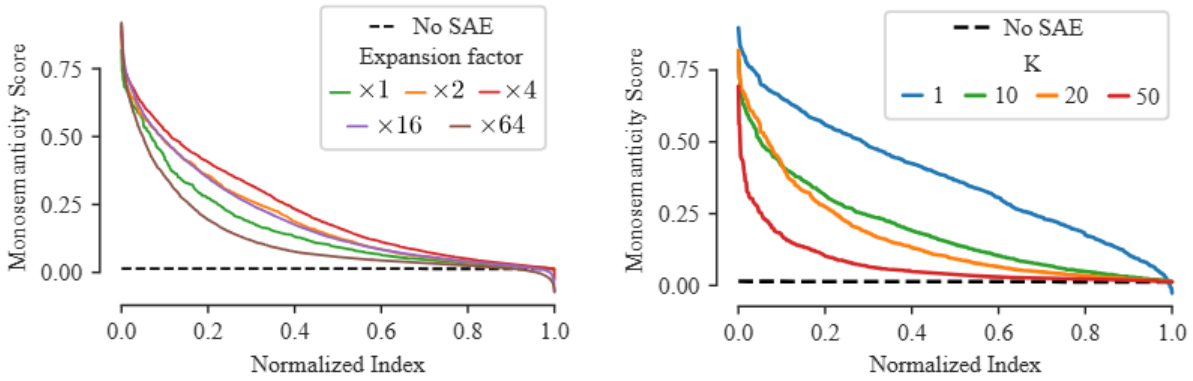
Table 1: Sparse Autoencoders (SAEs) decompose "No SAE" neurons into more monosemantic units as shown by MonoSemanticity (MS) score. Higher SAE expansion factors yield higher MS scores.

(a) Highest MS scores of neurons in various CLIP ViT-Large [47] layers.

SAE type	Layer	No SAE	Expansion factor ×1	×2	×4	×8	×16	×64
BatchTopK [8]	11	0.01	0.61	0.73	0.71	0.87	0.90	1.00
	17	0.01	0.65	0.79	0.86	0.86	0.93	1.00
	22	0.01	0.66	0.79	0.80	0.88	0.92	1.00
	23	0.01	0.73	0.72	0.83	0.89	0.93	1.00
	last	0.01	0.57	0.78	0.78	0.81	0.85	1.00
Matryoshka [9, 39]	11	0.01	0.84	0.90	0.95	1.00	0.89	1.00
	17	0.01	0.86	0.84	0.93	0.94	0.96	1.00
	22	0.01	0.83	0.83	0.87	0.94	1.00	1.00
	23	0.01	0.82	0.84	0.89	0.93	0.96	1.00
	last	0.01	0.82	0.91	0.89	0.93	0.91	1.00

(b) Top MS scores of neurons from last layers of different vision encoders. Improvements in MS score from applying Matryoshka SAEs are consistent across all the models.

Vision Encoder	No SAE	Exp. factor ×1	×4
WebSSL [18]	0.01	0.79	0.92
CLIP [47]	0.01	0.82	0.89
SigLIP [61]	0.01	0.83	0.88
AIMv2 [19]	0.01	0.59	0.85



(a) Impact of expansion factor ε (b) Impact of sparsity factor K

Figure 5: MonoSemanticity scores in decreasing order across neurons, normalized by width. Results are shown for the last layer of the model, without SAE ("No SAE", in black dashed line), and with SAE in straight lines using either (a) different expansion factors ($\varepsilon = 1$, $\varepsilon = 2$, $\varepsilon = 4$, $\varepsilon = 16$, $\varepsilon = 64$) or (b) different sparsity levels ($K = 1$, $K = 10$, $K = 20$, and $K = 50$).

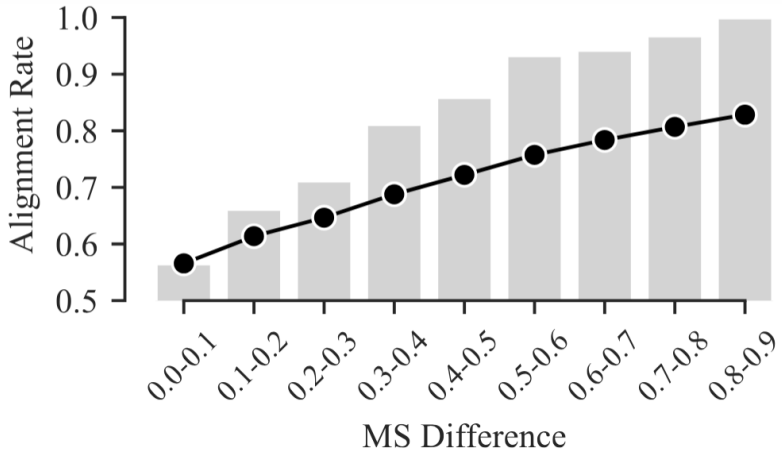


Figure 4: Alignment Rate (AR, %) of humans with MS score when judging which neuron in a pair is more monosemantic, grouped by MS difference between the neurons. Bars show AR per interval; dots show cumulative AR up to that interval.

Evaluating Interpretability of VLM Neurons

Table 2: Percentage of generations meeting evaluation criteria for concept insertion and suppression. SAE-derived steering directions yield higher success rates than Difference-in-Means (DiffMean) [2].

(a) Concept insertion

	Ours	DiffMean
Desired concept appeared	48.7	**53.1**
Base prompt followed	**85.8**	66.2
Both criteria satisfied	**42.4**	35.8

(b) Concept suppression

	Ours	DiffMean
Desired concept removed	**64.4**	64.0
Unrelated concept kept	**81.4**	38.7
Both criteria satisfied	**52.5**	33.3

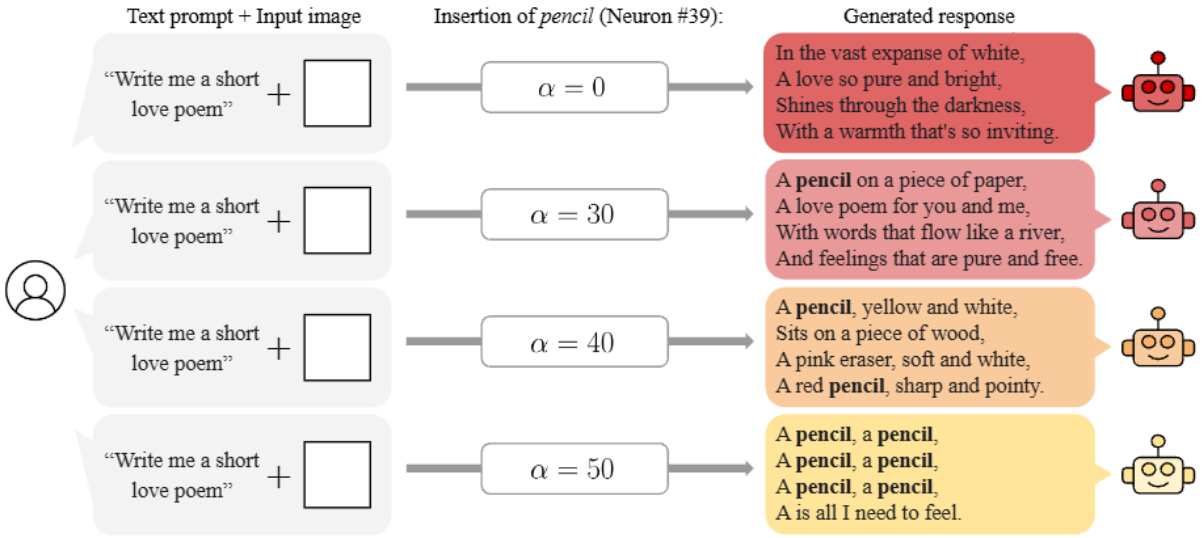


Figure 6: Steered LLaVA outputs after clamping the activation values of a chosen neuron, i.e. Neuron #39 of SAE in CLIP corresponding to *pencil*. Initially, the poem follows the given instruction (the prompt and white image), but as the intervention weight α increases, it becomes increasingly influenced by the neuron's concept, first mentioning the *pencil*'s attributes, then the *pencil* itself. This shows that our interventions enable new capabilities for the unsupervised steering of these models.

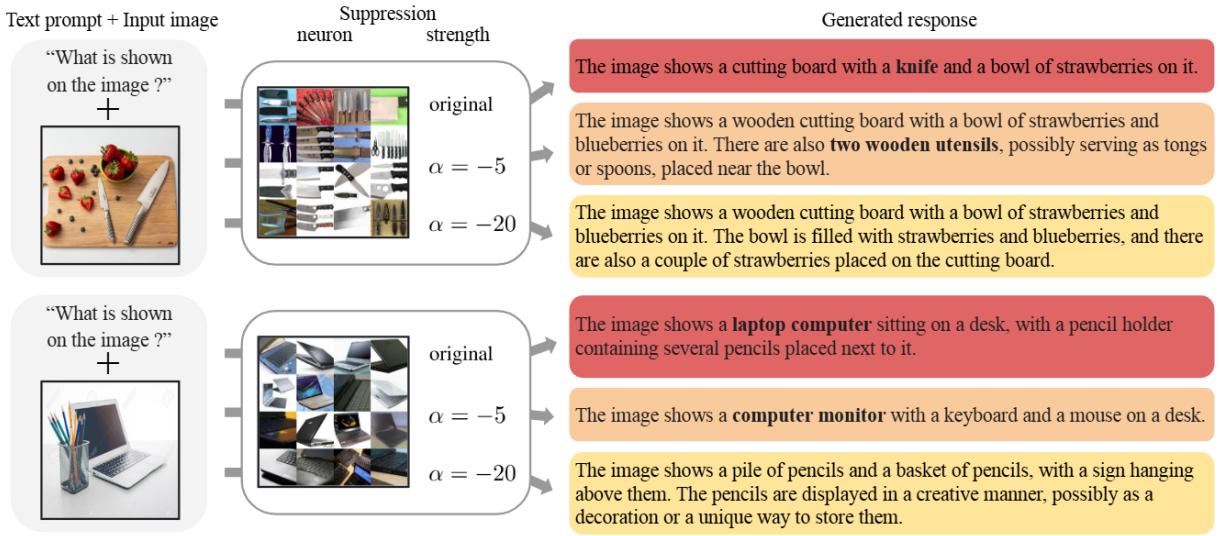


Figure 7: Overriding neuron activations with negative values allows concept suppressing. Originally, LLaVA describes the images by mentioning all the objects visible on the input images. However, as we decrease the value α of neurons associated with *knives* and *laptops*, it first confuses them with wooden utensils and computer monitor, then eventually ignores them completely. At the same time, it continues to faithfully describe other objects like wooden board, strawberries, and pencil holder.

Conclusion

- MonoSemanticity score를 도입하였으며 Matryoshka SAE가 일반 BatchTopK보다 좋은 성능을 보임을 확인함.

- 추후 Large-scale human study를 통해 사람의 인식과 MS의 alignment를 확인할 예정임.

- **Limitations**
  - **SAE 중심 분석의 한계**: SAE 기반 뉴런 분석에 편중되어 있음.
  - **Fine-grained Steering의 한계**: MS 점수가 높더라도 세밀한 개념 조종이 보장되지는 않음.
    - '골든 리트리버' 뉴런을 자극해도 구체적인 품종 대신 일반적인 '개(Dog)' 관련 단어가 출력될 수 있음.
    - LLM의 세부 개념 이해 부족 및 비전-언어 간 정렬 불완전성에 기인함.
  - **Steering 효과가 없는 뉴런 존재**: 시각적 특징을 잘 감지하더라도 뉴런 활성값 조작이 최종 텍스트 출력에 영향을 주지 않는 경우가 존재함.